VAE – Latent Representation

β –VAE: motivation

- Tries to disentangle the latent representations
- What does it mean?
  - Each factor $z^{(i)}$ is uncorrelated with other factors
- Why is this good?
  - Better interpretability of the latent space
- How to do this?
  - Remember the prior P(**z**) had a diagonal covariance
  - We can enforce that more strongly

Higgins et al. "**β-VAE: LEARNING BASIC VISUAL CONCEPTS WITH A CONSTRAINED VARIATIONAL FRAMEWORK**", ICLR 2017

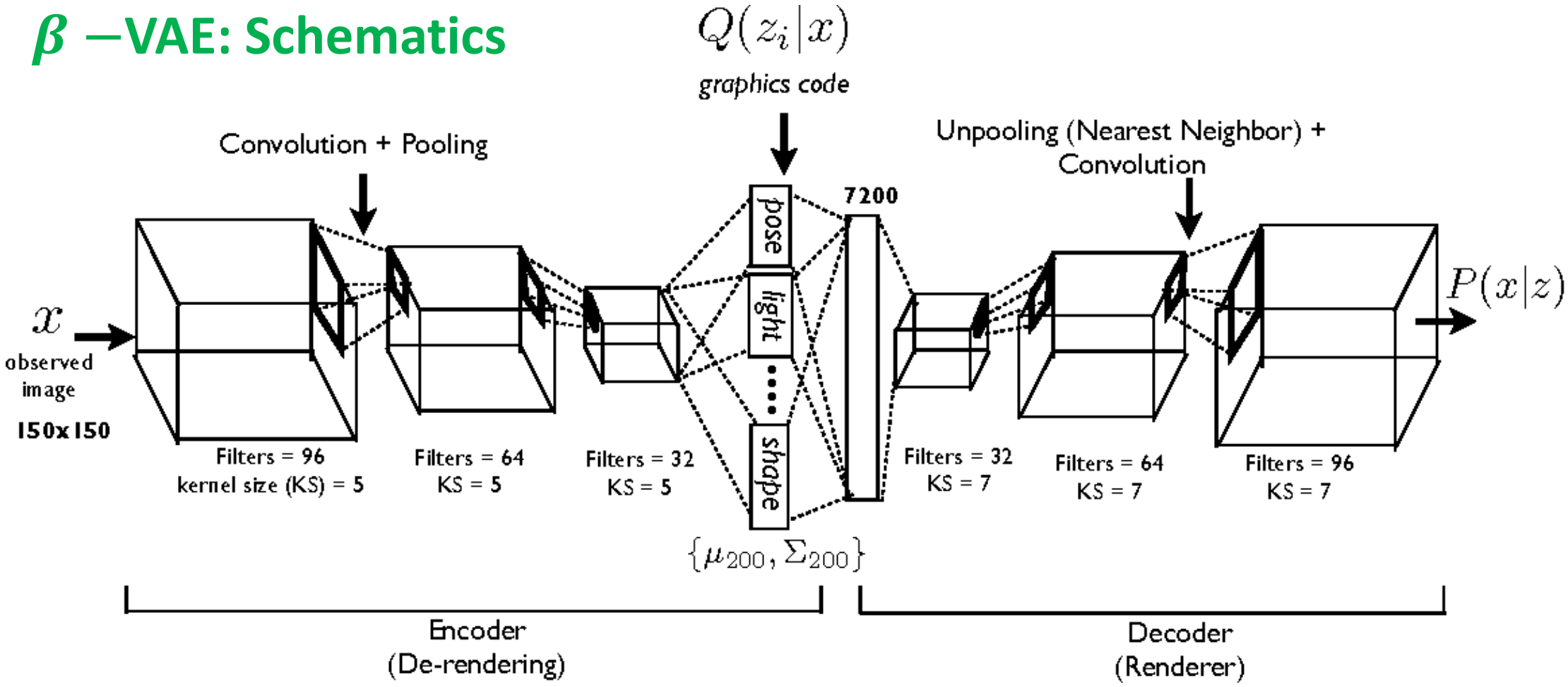
β –VAE: Training Equation

$$\max_{\{\boldsymbol{\omega}_i\}} \sum_{i=1}^{n} \left(\mathbb{E}_{Q_{\boldsymbol{\omega}_i}(\mathbf{z}|\mathbf{x}_i)} \log P(\mathbf{x}_i|\mathbf{z}) - \beta D_{KL}\left(Q_{\boldsymbol{\omega}_i}(\mathbf{z}|\mathbf{x}_i) || P(\mathbf{z}) \right) \right)$$

Higgins et al. "**β-VAE: LEARNING BASIC VISUAL CONCEPTS WITH A CONSTRAINED VARIATIONAL FRAMEWORK**", ICLR 2017

VAE – Latent Representation

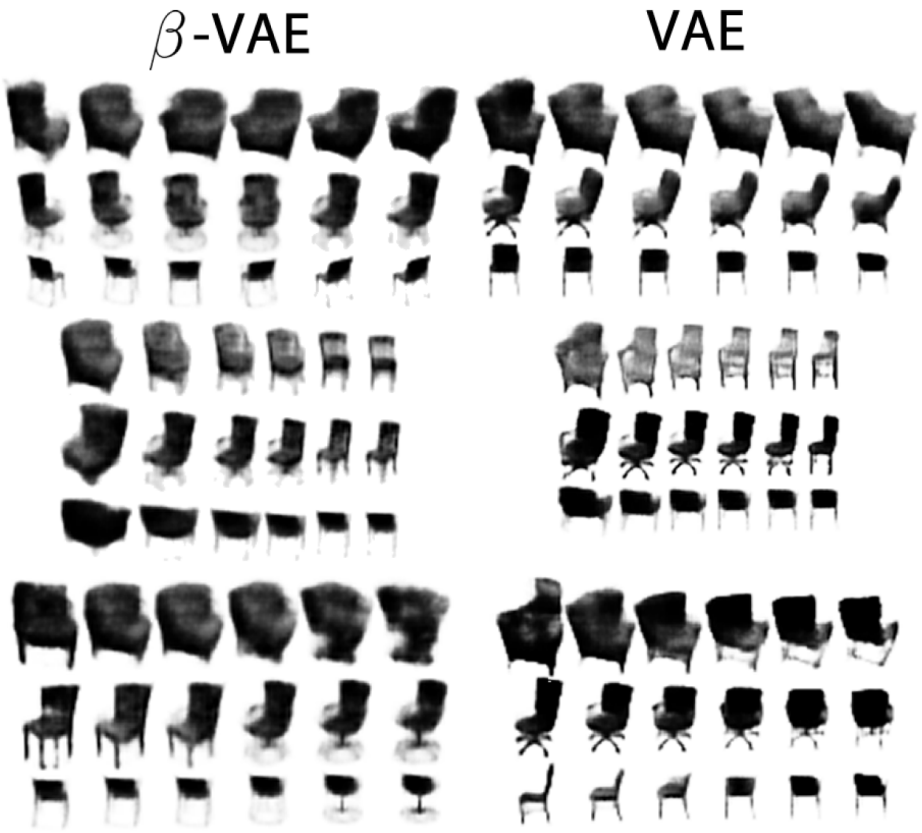
β –VAE: Schematics



Higgins et al. "**β-VAE: LEARNING BASIC VISUAL CONCEPTS WITH A CONSTRAINED VARIATIONAL FRAMEWORK**", ICLR 2017

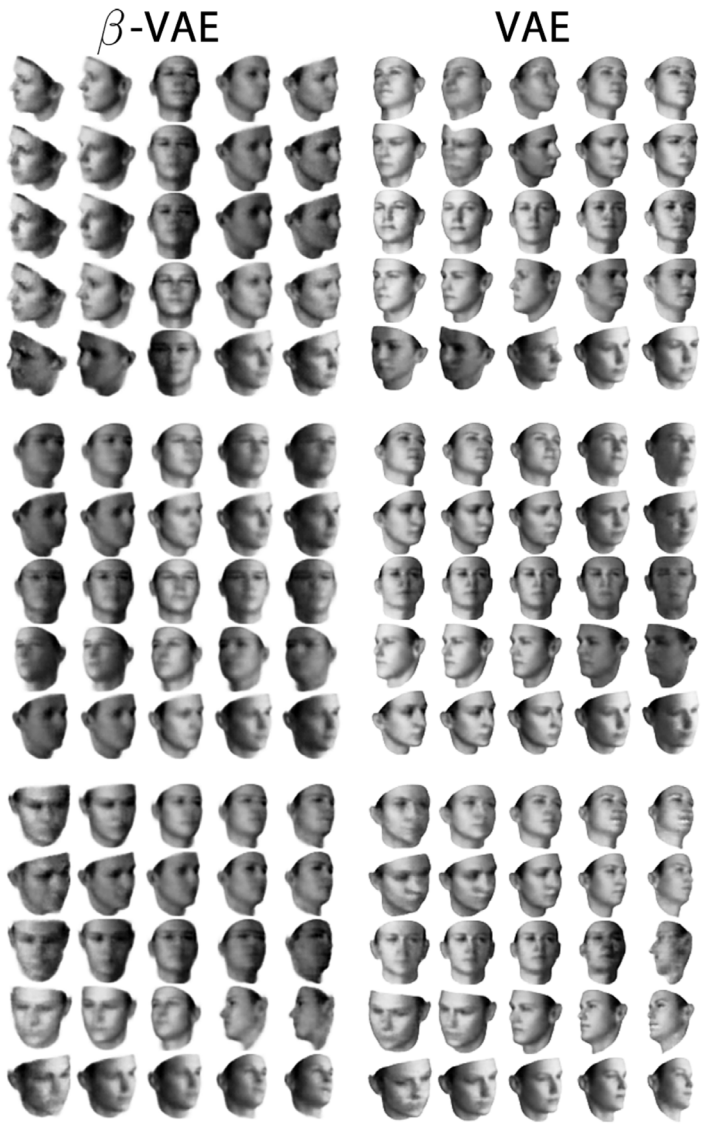
VAE – Latent Representation

β –VAE: Results



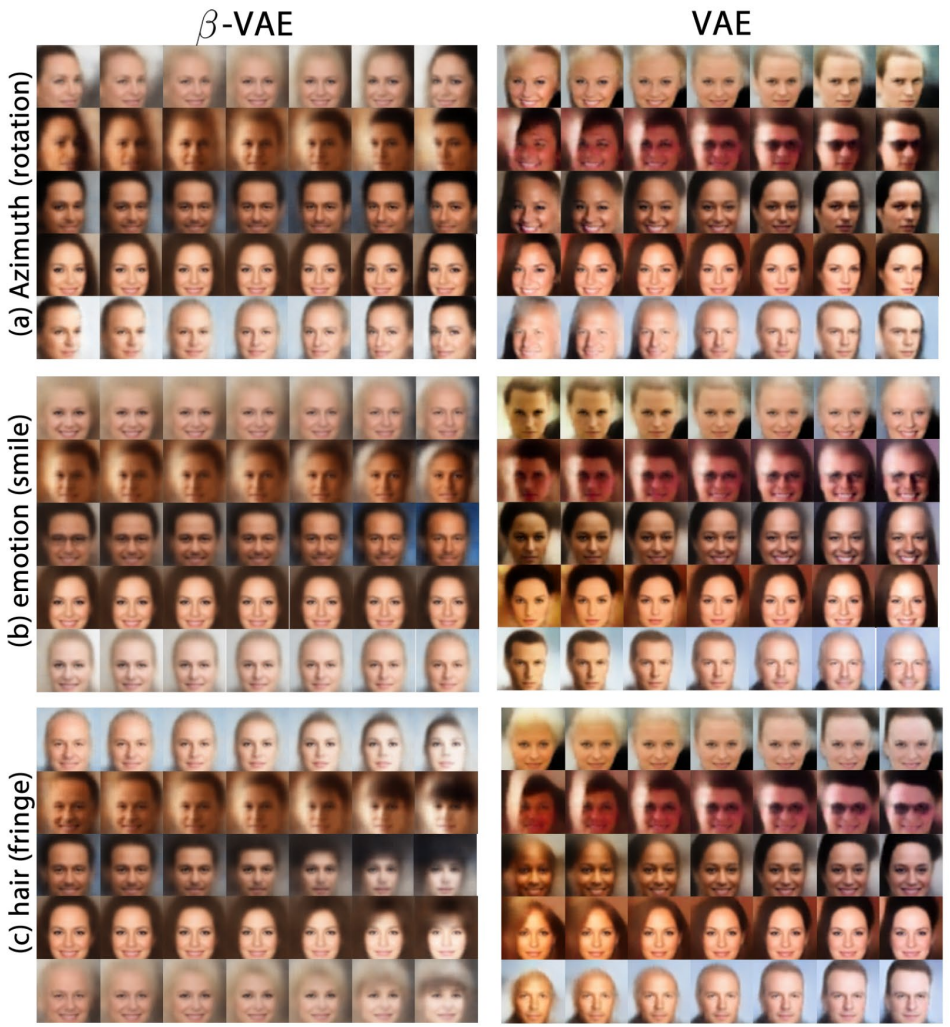
VAE – Latent Representation

β –VAE: Results



VAE – Latent Representation

β –VAE: Results

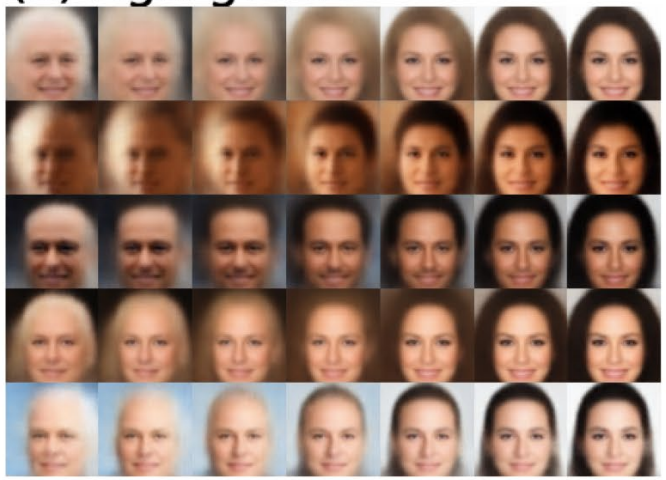


β –VAE: Results

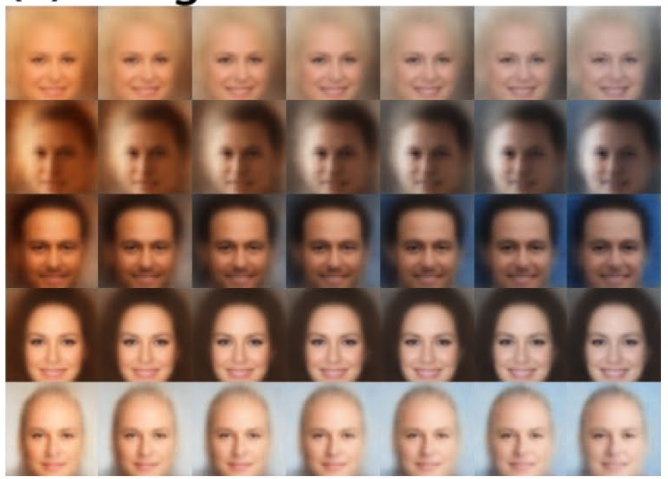
(a) Skin colour



(b) Age/gender

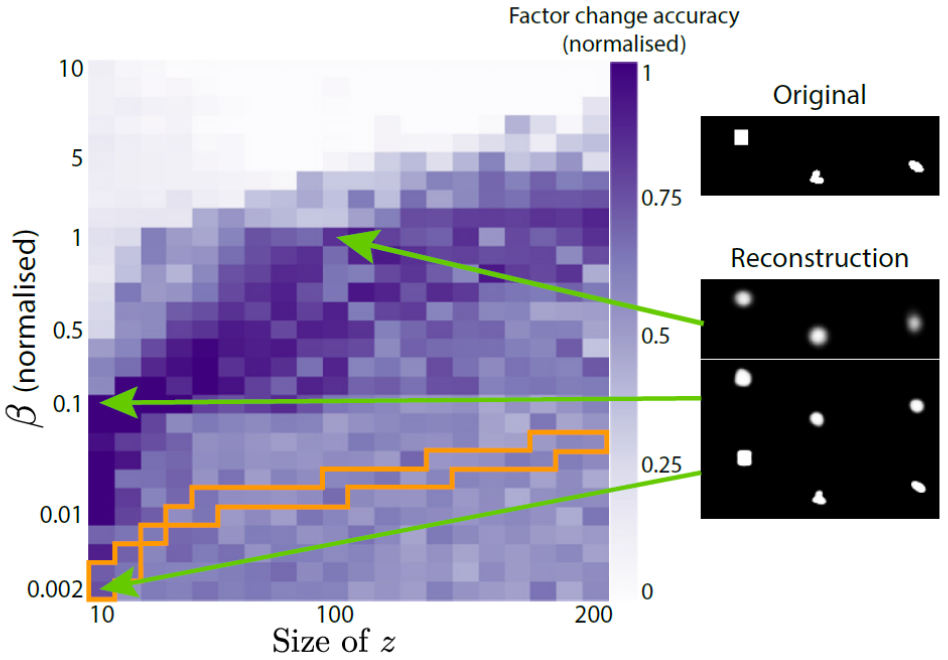


(c) Image saturation



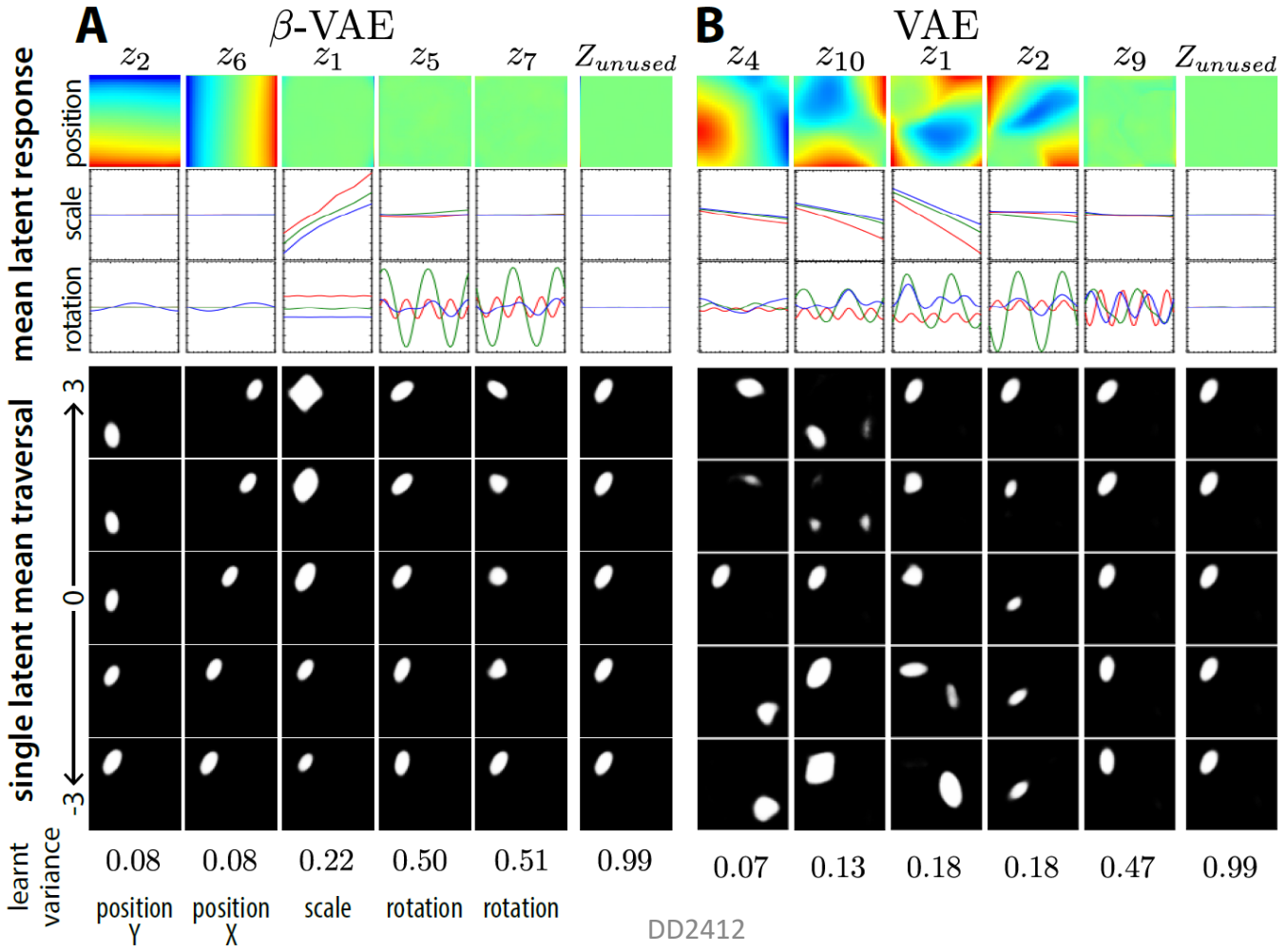
β –VAE: Results

Model	Disentanglement metric score
Ground truth	*100%*
Raw pixels	$45.75 \pm 0.8\%$
PCA	$84.9 \pm 0.4\%$
ICA	$42.03 \pm 10.6\%$
DC-IGN	$\mathbf{99.3 \pm 0.1\%}\%$
InfoGAN	$73.5 \pm 0.9\%$
VAE untrained	$44.14 \pm 2.5\%$
VAE	$61.58 \pm 0.5\%$
β-VAE	$\mathbf{99.23 \pm 0.1\%}$



VAE – Latent Representation

β –VAE: Results



VAE - summary

- Simple implementation and stable training procedure
- No overwhelming hyper-parameters
- Very good for encoding samples
- Good for understanding the underlying generative factors of data (using tricks like beta-VAE)
- Good for generative modeling (slightly blurry images though)
- OK for density estimation

Check out [Tschannen, Bachem, Lucic, “Recent Advances in Autoencoder-Based Representation Learning”, NIPS 2018]